OVER A CATENARY NOETHERIAN LOCAL NORMAL DOMAIN EVERY RADICAL IDEAL CONTAINS A NONZERO ELEMENT WHOSE QUOTIENT IS REDUCED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span includes the author's own \begin{reference} block, in which the author notes that the cited work 'has this result without the assumption that the ring is catenary' and cites it; that block is delivered verbatim. Because the project defines the reference environment as a comment environment, the citation is typeset as a comment and contributes no unresolved citation. The author's typos in the counted proof span ('q_1 cap ... cap q_s' for one occurrence of the localizations, and the index j' used in the display defining c) are reproduced unchanged.

Allowed standard prerequisites (editorial).

- Catenary local rings: every height one prime has dimension d - 1, and normality giving R = intersection of the localizations at height one primes. used at proof lines 38405-38407 and invoked through the cited Algebra lemmas at proof lines 38432-38434. Makes Spec(R/(f + h)) equidimensional and lets reducedness be tested on the height one primes.
- Discrete valuation rings at height one primes of a normal domain and the divisor of an element. used at proof lines 38414-38426. Supplies the valuations ord_{q_j} whose values m_j are exactly what must be shown to be 1.
- The height-one-polynomial bound on the number of minimal primes. invoked at proof lines 38437-38438 and applied at proof lines 38446-38455. Is the input that bounds t(f + h) uniformly, enabling the maximality argument.
- Prime avoidance and the second symbolic power of a height one prime. used at proof lines 38487-38500. Produces the elements a_j and b_j from which the correcting element c is assembled.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00PD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 927–927. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 031S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3995–3995. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00IX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 695–695. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00KV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 765–765. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 031T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3996–3996. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00DS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 508–508. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0311: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3968–3968. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 38016: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38016–38016. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 38297: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38297–38297. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 38138: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38138–38138. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 38108: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38108–38108. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 38348: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38348–38348. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Proposition 0.2. *Let R be a catenary Noetherian local normal domain. Let $J \subset R$ be a radical ideal. Then there exists a nonzero element $f \in J$ such that R/fR is reduced.*

Proof. The proof is the same as that of Lemma 38138, using Lemma 38297 instead of Lemma 38016 and Lemma 38348 instead of Lemma 38108. We can use Lemma 38297 because R is a catenary domain, so every height one prime ideal of R has dimension $d-1$, and hence the spectrum of $R/(f+h)$ is equidimensional. For the convenience of the reader we write out the details.

Let $f \in J$ be a nonzero element. We will modify f slightly to obtain an element that generates a radical ideal. The localization $R_{\mathfrak{p}}$ of R at each height one prime ideal $\mathfrak{p}$ is a discrete valuation ring, see Algebra, Lemma 00PD or Algebra, Lemma 031S. We denote by $\mathrm{ord}_{\mathfrak{p}}(f)$ the corresponding valuation of f in $R_{\mathfrak{p}}$. Let $\mathfrak{q}_1, \ldots, \mathfrak{q}_s$ be the distinct height one prime ideals containing f. Write $\mathrm{ord}_{\mathfrak{q}_j}(f) = m_j \geq 1$ for each j. Then we define $\mathrm{div}(f) = \sum_{j=1}^{s} m_j \mathfrak{q}_j$ as a formal linear combination of height one primes with integer coefficients. The ring R/fR is reduced if and only if $m_j = 1$ for $j = 1, \ldots, s$. Namely, if m_j is 1 then $(R/fR)\mathfrak{q}_j$ is reduced and $R/fR \subset \prod (R/fR)_{\mathfrak{q}_j}$ as $\mathfrak{q}_1, \ldots, \mathfrak{q}_j$ are the associated primes of R/fR, see Algebra, Lemmas 0311 and 031T.

Choose and fix $g_2, \ldots, g_{d-1}$ and N as in Lemma 38348. For a nonzero $y \in R$ denote $t(y)$ the number of primes minimal over y. Since R is a normal domain, these primes are height one and correspond 1-to-1 to the minimal primes of R/yR (Algebra, Lemmas 00KV and 031T). For example $t(f) = s$ is the number of primes $\mathfrak{q}_j$ occurring in $\mathrm{div}(f)$. Let $h \in \mathfrak{m}^N$. Because R is catenary, for each height one prime $\mathfrak{p}$ of R we have $\dim(R/\mathfrak{p}) = d$. Hence by Lemma 38297 we have

$$\begin{aligned} t(f+h) &\leq \mathrm{length}_{R/(f+h)}(R/(f+h, g_1, \ldots, g_{d-1})) \\ &= \mathrm{length}_R(R/(f+h, g_1, \ldots, g_{d-1})) \\ &= \mathrm{length}_R(R/(f, g_1, \ldots, g_{d-1})) \end{aligned}$$

see Algebra, Lemma 00IX for the first equality. Therefore we see that $t(f+h)$ is bounded independent of $h \in \mathfrak{m}^N$.

By the boundedness proved above we may pick $h \in \mathfrak{m}^N \cap J$ such that $t(f+h)$ is maximal among such h. Set $f' = f + h$. Given $h' \in \mathfrak{m}^N \cap J$ we see that the number $t(f' + h') \leq t(f+h)$. Thus after replacing f by f' we may assume that for every $h \in \mathfrak{m}^N \cap J$ we have $t(f+h) \leq s$.

Next, assume that we can find an element $h \in \mathfrak{m}^N \cap J$ such that for each j we have $\mathrm{ord}_{\mathfrak{q}_j}(h) \geq 1$ and $\mathrm{ord}_{\mathfrak{q}_j}(h) = 1 \Leftrightarrow m_j > 1$. Then $\mathrm{ord}_{\mathfrak{q}_j}(f+h) = 1$ for every j by elementary properties of valuations. Thus

$$\mathrm{div}(f+h) = \sum\nolimits_{j=1}^{s} \mathfrak{q}_j + \sum\nolimits_{k=1}^{v} e_k \mathfrak{r}_k$$

for some pairwise distinct height one prime ideals $\mathfrak{r}_1, \ldots, \mathfrak{r}_v$ and $e_k \geq 1$. However, since $s = t(f) \geq t(f+h)$ we see that $v = 0$ and we have found the desired element.

Now we will pick h that satisfies the above criteria. By prime avoidance (Algebra, Lemma 00DS) for each $1 \leq j \leq s$ we can find an element $a_j \in \mathfrak{q}_j \cap J$ such that $a_j \notin \mathfrak{q}_{j'}$ for $j' \neq j$. Next, we can pick $b_j \in J \cap \mathfrak{q}_1 \cap \ldots \cap q_s$ with $b_j \notin \mathfrak{q}_j^{(2)}$. Here $\mathfrak{q}_j^{(2)} = \{x \in R \mid \mathrm{ord}_{\mathfrak{q}_j}(x) \geq 2\}$ is the second symbolic power of $\mathfrak{q}_j$. Prime avoidance applies because the ideal $J' = J \cap \mathfrak{q}_1 \cap \ldots \cap q_s$ is radical, hence R/J' is reduced, hence $(R/J')_{\mathfrak{q}_j}$ is reduced, hence J' contains an element x with $\mathrm{ord}_{\mathfrak{q}_j}(x) = 1$, hence

$J' \not\subset \mathfrak{q}_j^{(2)}$. Then the element

$$c = \sum\nolimits_{j=1,\ldots,s} b_j \times \prod\nolimits_{j' \neq j} a_{j'}$$

is an element of J with $\text{ord}_{\mathfrak{q}_j}(c) = 1$ for all $j = 1, \ldots, s$ by elementary properties of valuations. Finally, we let

$$h = c \times \prod\nolimits_{m_j=1} a_j \times y$$

where $y \in \mathfrak{m}^N$ is an element which is not contained in $\mathfrak{q}_j$ for all j. $\square$